Decompose a Goal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected function-and-call evidence record.

Match parameters, trace calls, repair constants and define the independent stock-count function.

Keep one habit: Keep definitions, supplied arguments, expected results and actual output/error together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the re**call** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Precision in Algorithms

MAIN LESSON · SECTION 12

W1 · Match arguments and parameters

> Use the supplied complete function definition.

Your evidence record

- In def show_remaining(capacity, used), identify the **function** name and **parameter**s. In show_remaining(15, 6), identify the **argument**s and their matching **parameter**s.

```
def show_remaining(capacity, used):
    remaining = capacity - used
    print(remaining)

show_remaining(12, 4)
show_remaining(10, 7)
```

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace calls and output order

> Predict before running; a definition alone does not print.

Your evidence record

- Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair fixed-number calculations

> Show which arguments the faulty body ignored.

Your evidence record

- The faulty body calculates 12 - 4 regardless of the **call**. State the **output** and expected **output** for show_remaining(11, 2). Repair the expression and name an earlier **call** to rerun.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Inspect the input contract

> Successful subtraction does not establish a valid cupboard record.

Your evidence record

- The correct subtraction **function** prints -1 for show_remaining(4, 5). Does successful execution make this a valid cupboard record? Explain what the **function** assumes and what needs checking.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Create a new function independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Create a new function independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Separate reader profile, book data, score calculation, ranking, and explanation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Follow the execution order for **Function**s and **argument**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A class wants to recommend one of five library books.

Decompose the goal into checkable steps.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 2 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

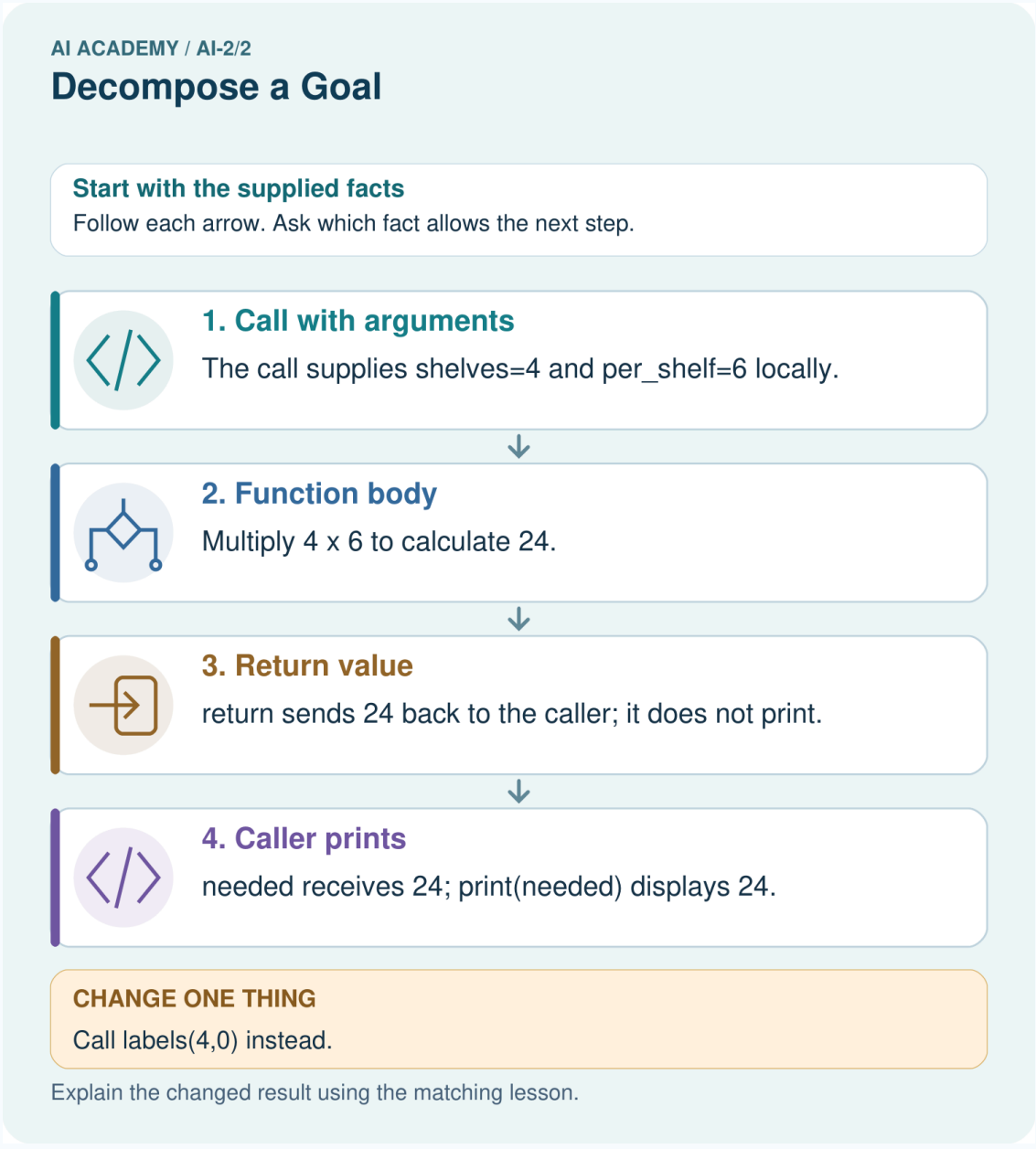
> Keep the explanation and the support you used.

Use this when helpful

A fictional library computes total spine labels needed for several equal shelves.

Define labels(shelves,per_shelf): return shelves*per_shelf.

Call needed=labels(4,6), then print(needed). No other side effects are defined.



Follow the labelled picture with the matching explanation. Open the vector diagram

What happens visibly if the **call**er stores needed but omits print(needed)?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional library computes total spine labels needed for several equal shelves.
- Which labelled step matches this detail? needed receives 24; print(needed) displays 24.
- 1. **Function** body
- 2. **Call**er prints
- 3. **Call** with **argument**s

- 4. Return value

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: **Call**er prints needed receives 24; print(needed) displays 24. A **function** separates one named job from the surrounding program.

Arguments supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Changed-case question

- Changed-case practice: **Call** labels(4,0) instead. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. The returned result is 0, and the **call**er prints 0.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The returned result is 0, and the **call**er prints 0. Four shelves times zero labels per shelf is zero.

Repair and save your work

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

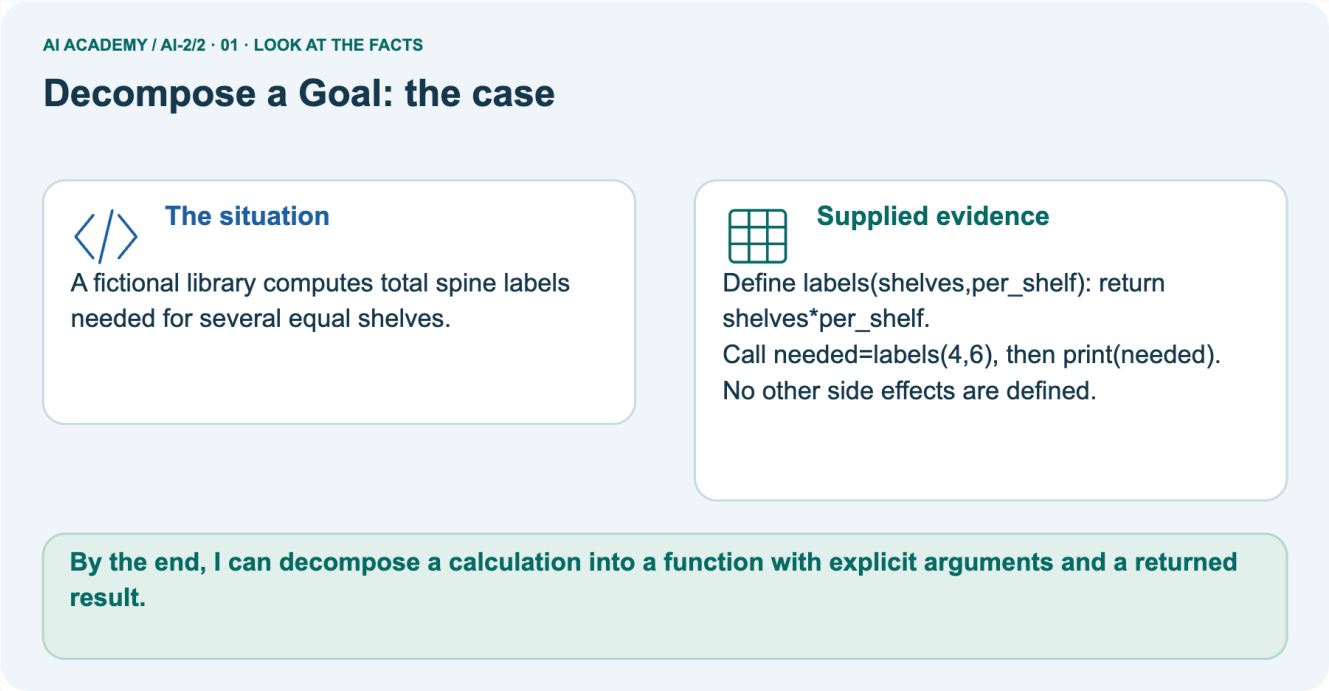
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Decompose a Goal: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

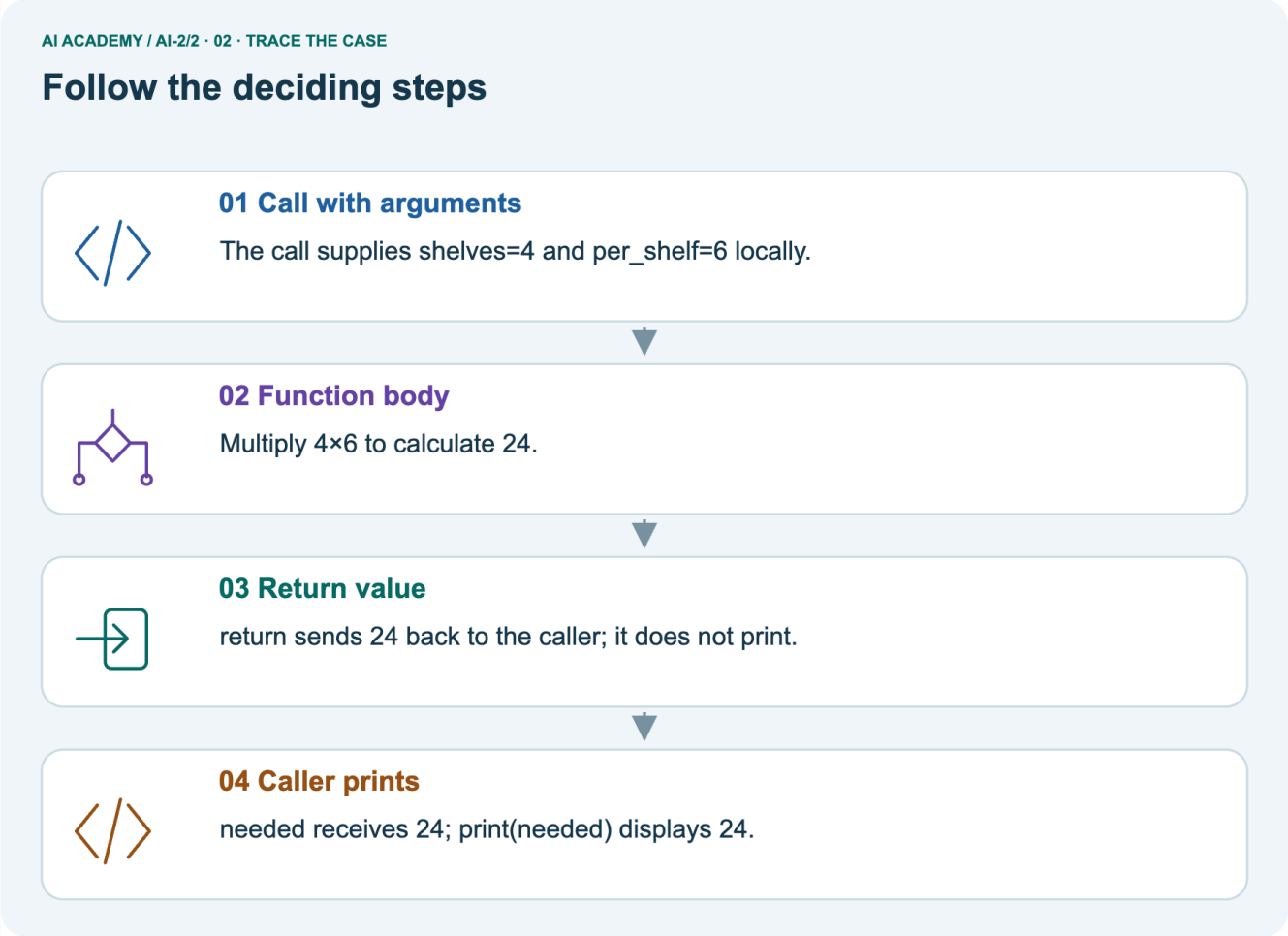
A fictional library computes total spine labels needed for several equal shelves. Supplied facts: Define labels(shelves,per_shelf): return shelves*per_shelf. **Call** needed=labels(4,6), then print(needed).

No other side effects are defined.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Call with **argument**s: The **call** supplies shelves=4 and per_shelf=6 lo**call**y. → **Function** body: Multiply 4×6 to calculate 24. → Return value: return sends 24 back to the **call**er; it does not print. → **Call**er prints: needed receives 24; print(needed) displays 24.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Arguments matched in order

AI ACADEMY / AI-2/2 · 03 · INSPECT THE LESSON MODEL

Arguments matched in order

Call	Parameter values	Printed output
show_remaining(12, 4)	capacity 12; used 4	8
show_remaining(10, 7)	capacity 10; used 7	3

Each call runs the same body with the values supplied for that call.

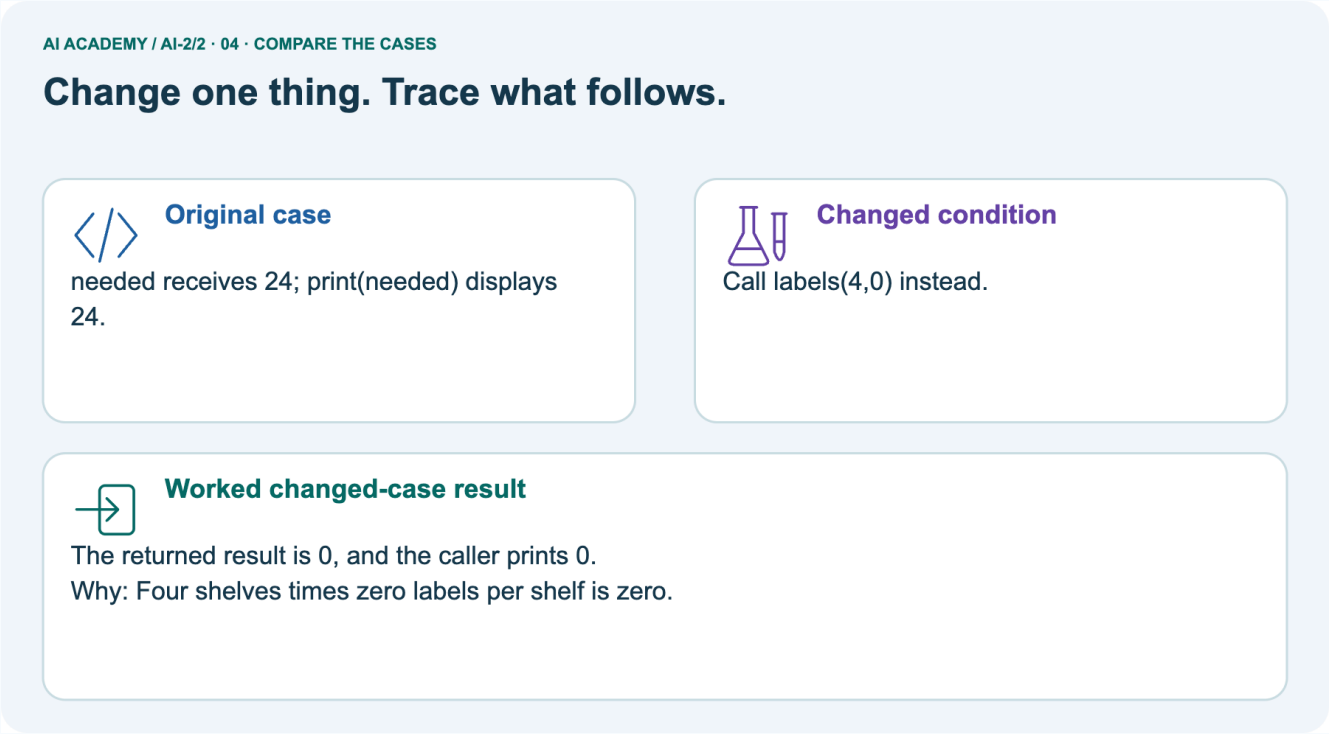
Follow the labelled picture with the matching explanation. Open the vector diagram

Arguments matched in order. show_remaining(12, 4) | capacity 12; used 4 | 8. show_remaining(10, 7) | capacity 10; used 7 | 3

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

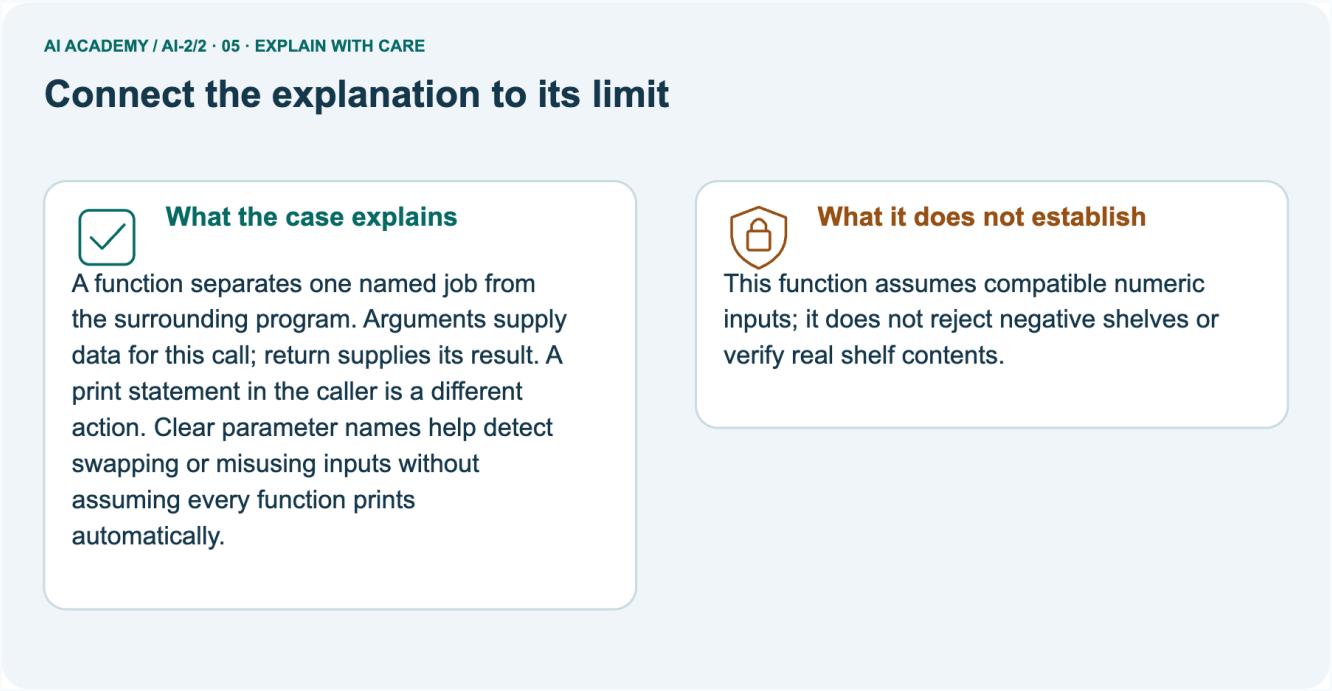
Original case: needed receives 24; print(needed) displays 24. Changed condition: **Call** labels(4,0) instead. Expected result: The returned result is 0, and the **call**er prints 0.

Reason: Four shelves times zero labels per shelf is zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

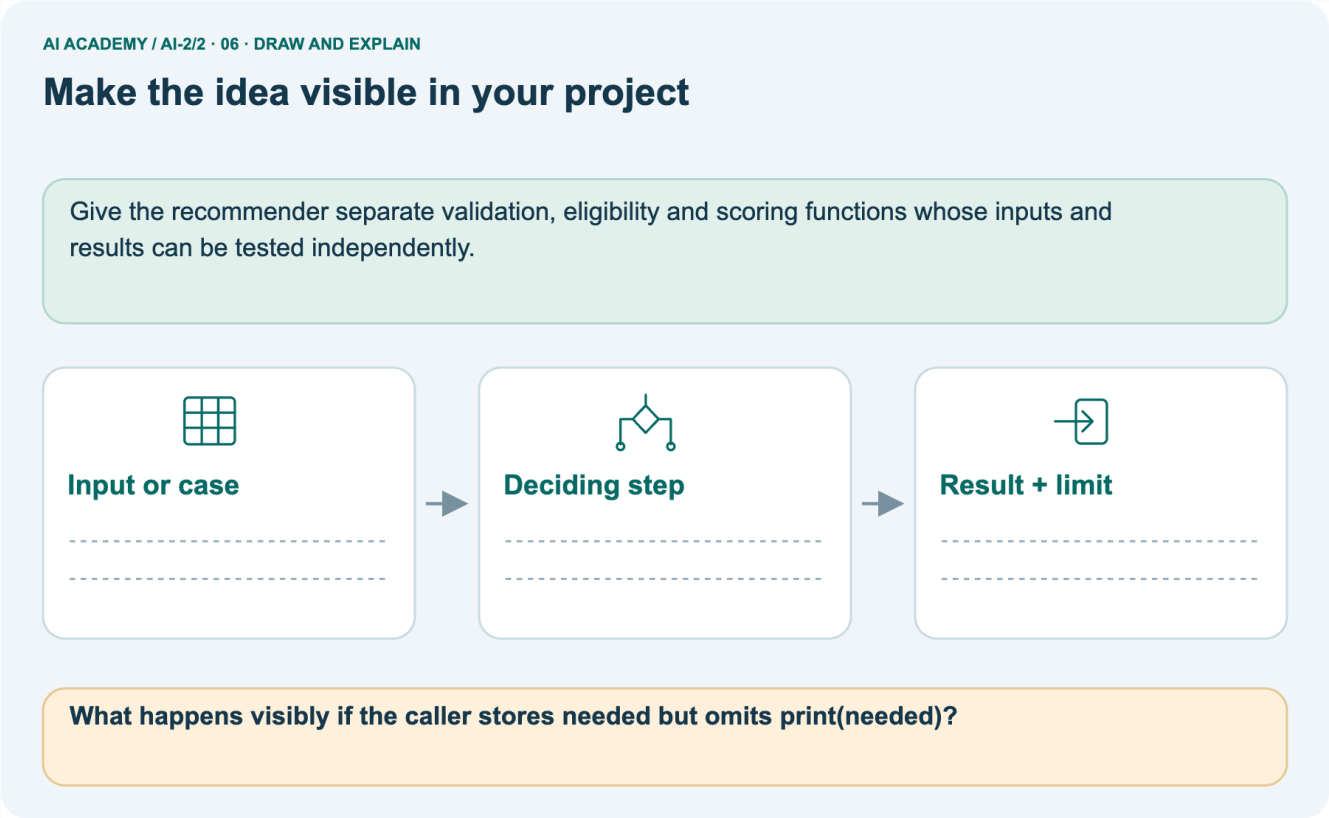
Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Limit: This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently. Draw your **input**, deciding step, result and limit.

Supported question: What happens visibly if the **call**er stores needed but omits print(needed)?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Decompose a Goal: the case

02. Follow the deciding steps

03. Arguments matched in order

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.